THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 21	Square first, then multiply.
2	**(a)** $2x + 13$	Expand each bracket, watching the signs.
3	**(c)** $16x + 6$	Perimeter $= 2\times$(length + breadth).
4	**(b)** 55	Substitute, multiply, then halve.
5	**(c)** $2n + 4$	Test the expressions with small values of n.
6	**(b)** 22	Substitute carefully; xy means $x \times y$.
7	**(d)** 23	Apply the rule to the input.
8	**(d)** 77	$x^2 = x \times x$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Three times n is 3n, and five more is 3n + 5.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	2 x 4 + 3 = 11. Both are true, but R does not explain the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Expand: $7x - 14 - 3x + 15$ (1); Answer: $4x + 1$ (1).	1+1
12	$2n + 2(n + 1) + 2(n + 2)$ (1); $= 6n + 6 = 6(n + 1)$ (1).	1+1
13	Substitute: $2(5)^2 - 3(5) + 4$ (1); $= 39$ (1).	1+1
14	Expand: $9x - 27 - 4x + 8$ (1); Answer: $5x - 19$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P = 2[(5x + 4) + (3x - 1)]$; $= 2(8x + 3)$ (1); $= 16x + 6$ (1); At $x = 6$: 102 cm (1).	1+1+1
16	Numbers: $n, n + 1, \ldots, n + 2$ (1); Sum $= 3n + 3$ (1); At $n = 7$: 24 (1).	1+1+1
17	A: $20 + 15k$ and B: $60 + 10k$ (1); For 6 km: A $= 110$ and B $= 120$ (1); Taxi A is cheaper by ₹10 (1).	1+1+1
	OR: $P = 2[(7x + 3) + (2x - 1)]$; $= 2(9x + 2)$ (1); $= 18x + 4$ (1); At $x = 5$: 94 cm (1).	1+1+1
18	Numbers: $n, n + 1, \ldots, n + 3$ (1); Sum $= 4n + 6$ (1); At $n = 5$: 26 (1).	1+1+1
	OR: A: $30 + 12k$ and B: $90 + 7k$ (1); For 15 km: A $= 210$ and B $= 195$ (1); Taxi B is cheaper by ₹15 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 \times 5 + 1 = 11$ sticks (1). (ii) $2n + 1$ (1). (iii) $2 \times 10 + 1 = 21$ sticks (1). (iv) $2n + 1 = 41$ gives $2n = 40$, so $n = 20$ triangles (2).	1+1+1+2
20	(i) $5(12) + 2 = 62$ (1); (ii) Figure 1 $= 7$ and figure 2 $= 12$ (1); Total $= 19$ (1); (iii) $62 - 57 = 5$ more tiles (2).	1+1+1+2
	OR: Brother's money $= 2x + 20$ (1); Total $= x + (2x + 20) = 3x + 20$ (1); At $x = 35$: brother $=$ ₹90 (1); Total $=$ ₹125 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $18x + 6$	Perimeter $= 2\times$(length + breadth).
2	**(c)** 105	Substitute, multiply, then halve.
3	**(b)** $4n + 3$	Test the expressions with small values of n.
4	**(b)** 9	Substitute carefully; xy means $x \times y$.
5	**(b)** 29	Apply the rule to the input.
6	**(a)** 79	$x^2 = x \times x$.
7	**(b)** 32	Square first, then multiply.
8	**(c)** $4x + 14$	Expand each bracket, watching the signs.
9	**(c)** A is true but R is false.	Like terms are added by adding their numbers: 5a. The letter part does not change, so R is false.
10	**(d)** A is false but R is true.	4x and 3y are unlike terms and cannot be combined, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5n + 5(n + 1) + 5(n + 2)$ (1); $= 15n + 15 = 15(n + 1)$ (1).	1+1
12	Substitute: $3(4)^2 - 5(4) + 1$ (1); $= 29$ (1).	1+1
13	Expand: $8x - 8 - 5x + 20$ (1); Answer: $3x + 12$ (1).	1+1
14	$4n + 4(n + 1) + 4(n + 2)$ (1); $= 12n + 12 = 12(n + 1)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P = 2[(4x + 6) + (5x - 2)]$; $= 2(9x + 4)$ (1); $= 18x + 8$ (1); At $x = 8$: 152 cm (1).	1+1+1
16	Numbers: $n, n + 1, \ldots, n + 4$ (1); Sum $= 5n + 10$ (1); At $n = 6$: 40 (1).	1+1+1
17	A: $40 + 13k$ and B: $100 + 8k$ (1); For 10 km: A $= 170$ and B $= 180$ (1); Taxi A is cheaper by ₹10 (1).	1+1+1
	OR: $P = 2[(6x + 2) + (4x - 3)]$; $= 2(10x - 1)$ (1); $= 20x - 2$ (1); At $x = 7$: 138 cm (1).	1+1+1
18	Numbers: $n, n + 1, \ldots, n + 5$ (1); Sum $= 6n + 15$ (1); At $n = 4$: 39 (1).	1+1+1
	OR: A: $25 + 14k$ and B: $65 + 9k$ (1); For 9 km: A $= 151$ and B $= 146$ (1); Taxi B is cheaper by ₹5 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $30 + 12 \times 5 =$ ₹90 (1). (ii) $12k + 30$ (1). (iii) $12 \times 10 + 30 =$ ₹150 (1). (iv) $12k + 30 = 138$ gives $12k = 108$, so $k = 9$ km (2).	1+1+1+2
20	(i) $3(30) + 7 = 97$ (1); (ii) Figure 1 $= 10$ and figure 2 $= 13$ (1); Total $= 23$ (1); (iii) $97 - 94 = 3$ more tiles (2).	1+1+1+2
	OR: Brother's money $= 2x + 30$ (1); Total $= x + (2x + 30) = 3x + 30$ (1); At $x = 25$: brother $=$ ₹80 (1); Total $=$ ₹105 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $10n + 5$	Test the expressions with small values of n.
2	**(b)** 6	Substitute carefully; xy means $x \times y$.
3	**(c)** 43	Apply the rule to the input.
4	**(b)** 197	$x^2 = x \times x$.
5	**(c)** 120	Square first, then multiply.
6	**(c)** $4x + 33$	Expand each bracket, watching the signs.
7	**(a)** $26x + 2$	Perimeter $= 2\times$(length + breadth).
8	**(a)** 325	Substitute, multiply, then halve.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Adding four equal sides gives 4s.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	7 x 5 = 35. Both true, but R does not explain the calculation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $4(3)^2 - 2(3) + 6$ (1); $= 36$ (1).	1+1
12	Expand: $6x - 24 - 2x + 14$ (1); Answer: $4x - 10$ (1).	1+1
13	$6n + 6(n + 1) + 6(n + 2)$ (1); $= 18n + 18 = 18(n + 1)$ (1).	1+1
14	Substitute: $5(2)^2 - 6(2) + 7$ (1); $= 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P = 2[(8x + 5) + (3x - 2)]$; $= 2(11x + 3)$ (1); $= 22x + 6$ (1); At $x = 4$: 94 cm (1).	1+1+1
16	Numbers: $n, n + 1, \ldots, n + 6$ (1); Sum $= 7n + 21$ (1); At $n = 3$: 42 (1).	1+1+1
17	A: $10 + 18k$ and B: $70 + 8k$ (1); For 8 km: A $= 154$ and B $= 134$ (1); Taxi B is cheaper by ₹20 (1).	1+1+1
	OR: $P = 2[(3x + 7) + (6x - 4)]$; $= 2(9x + 3)$ (1); $= 18x + 6$ (1); At $x = 9$: 168 cm (1).	1+1+1
18	Numbers: $n, n + 1, \ldots, n + 7$ (1); Sum $= 8n + 28$ (1); At $n = 2$: 44 (1).	1+1+1
	OR: A: $35 + 16k$ and B: $95 + 11k$ (1); For 14 km: A $= 259$ and B $= 249$ (1); Taxi B is cheaper by ₹10 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3r + 2$ (1). (ii) $3 \times 12 + 2 = 38$ years (1). (iii) $3r + 2 + 6 = 3r + 8$ (1). (iv) $r + (3r + 2) + (3r + 8) = 7r + 10$ (1); for $r = 12$: $7 \times 12 + 10 = 94$ years (1).	1+1+1+2
20	(i) $8(9) + 5 = 77$ (1); (ii) Figure 1 $= 13$ and figure 2 $= 21$ (1); Total $= 34$ (1); (iii) $77 - 69 = 8$ more tiles (2).	1+1+1+2
	OR: Brother's money $= 2x + 25$ (1); Total $= x + (2x + 25) = 3x + 25$ (1); At $x = 60$: brother $=$ ₹145 (1); Total $=$ ₹205 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.